MTH 201 **Work & Length & Surface Area**

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. A 3lb bucket containing 20lbs of water is hanging at the end of a 20ft rope that weighs 4oz/ft. The other end of the rope is attached to a pulley.

a) How much work is required to wind the length of rope onto the pulley?

b) How much work is required to wind half of the rope onto the pulley?

2. A trough is in the shade of a half drum (half a cylinder lying on its side). The length of the tank is 6 feet and the radius is 1 foot. Assuming it is full of water, find the work done in pumping the water to the top of the tank. (The density of the water is $62.4\, lbs/ft^3$)

3. Kyle jogs along the curve defined by $f(x)=\frac{2(x-1)^{3/2}}{3}$ from $x=1$ to $x=4$. Mallory jogs along the straight line connecting those two points. Kyle and Mallory both start from $x=1$ at the same time and Kyle jogs at a speed of $\frac{7}{\sqrt{3}}$ units/s. What is the speed at which Mallory must run so that she arrives at $(4, f(4))$ at the same time as Kyle?

4. Consider the curve defined by

$$y=\sin(x) \quad for \;\; 0 \le x \le 3.$$

a) Setup, but do not evaluate, an integral which represents the length of this curve.

b) Setup, but do not evaluate, an integral which represents the surface area of the solid obtained by rotating this curve about the x-axis.

c) Setup, but do not evaluate, an integral which represents the surface area of the solid obtained by rotating this curve about the y-axis.

5. Suppose $f(x)$ is a function with the following properties:

(i) $f'(x)$ is a differentiable function.

(ii) $f(x) \le 1$ for all values of x.

(iii) Over any interval, the area between the curve $y = \sqrt{2 - f(x)^2}$ and the x-axis is equal to the arclength of the curve $f(x)$.

(iv) $f(0) = 1$

a) Rewrite property (iii) using integral notation.

b) Use the Fundamental Theorem of Calculus (Part 1) to confirm that the integrands from part (a) are equal.

c) Use your answer from part (b) to help determine a formula for $\frac{dx}{dy}$.

d) Show that there are actually infinitely many possible functions $f(x)$ that satisfies all the stated properties.